

Noncitizen Populations and Citizen VAP Redistricting

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June 28, 2026

Abstract

Evenwel v. Abbott, 578 U.S. 54 (2016), held unanimously that states *may* use total population as the redistricting base, but explicitly left open whether states *must* use citizen voting-age population (citizen VAP) or may adopt it as an alternative. The question is not merely academic: the gap between total population and citizen VAP exceeds five percentage points in approximately 80 of the 435 congressional districts, and the systematic geographic pattern of noncitizen concentration makes the choice of population base a substantial determinant of partisan and urban-rural representation balance.

This paper examines the legal framework governing noncitizen-population redistricting at both the congressional and state legislative levels, the empirical magnitude of the total-population–citizen-VAP divergence in the high-immigration states of California, Texas, New York, and Florida, and the partisan and geographic consequences of switching from total population to citizen VAP as the redistricting base. We also describe BISECT’s `--population-source cvap` implementation, which uses the Census CVAP special tabulation to run the recursive bi-section algorithm under a citizen-VAP balance constraint, enabling systematic comparison of plans under the two population definitions.

The paper’s central empirical finding is that switching from total population to citizen VAP for congressional redistricting produces a consistently Republican-advantaging shift: urban districts with large noncitizen populations lose population credit, requiring them to absorb additional territory; suburban and exurban districts with small noncitizen populations gain relative standing. The effect is largest in California (where the citizen-VAP–total-population gap exceeds 8 percentage points in Los Angeles County tracts), Texas (South Texas border districts), New York (New York City boroughs), and Florida (Miami-Dade). We estimate that switching to citizen VAP would shift 15–20 congressional seats nationally from urban-Democratic to suburban-Republican lean over a sustained redistricting cycle.

The legal distinction between congressional and state legislative redistricting is critical: *Article I*’s command that Representatives be “apportioned among the several States according to their respective Numbers” has consistently been interpreted to require total-population apportionment at the federal level, insulating congressional redistricting from citizen-VAP mandates. State legislative redistricting, addressed by *Evenwel*, presents a different constitutional question. We analyze the arguments for and against citizen-VAP mandates in both contexts.

1 Introduction

The question of whose population counts for redistricting has divided legal scholars, advocates, and political actors for generations. The most contested version of this question concerns noncitizens: approximately 22 million noncitizen adults reside in the United States, concentrated in urban metropolitan areas in a small number of high-immigration states. Whether those noncitizens should be counted for redistricting—and if so, whether they should count toward the total population or be excluded in favor of a citizen-only denominator—determines how political representation is distributed between urban immigrant communities and suburban citizen-majority communities.

The Supreme Court addressed one dimension of this question in *Evenwel v. Abbott* [Evenwel v. Abbott, 2016]: states *may* use total population as their redistricting base, and the challengers' argument that the Equal Protection Clause *requires* citizen VAP was rejected unanimously. But the Court conspicuously declined to hold that total population is the *only* permissible base, leaving open whether states *may* use citizen VAP instead. A forthcoming legal battle—perhaps already underway in some state capitals—will contest the other side of *Evenwel*: whether the Constitution or federal statute *requires* citizen VAP, or at minimum *permits* it as an alternative.

This paper addresses four questions:

1. What is the legal framework governing the choice between total population and citizen VAP, separately for congressional and state legislative redistricting?
2. How large is the empirical divergence between total population and citizen VAP in the high-immigration states, and which districts are most affected?
3. What is the partisan direction of switching from total population to citizen VAP, and how many congressional seats would plausibly shift?
4. How does BISECT's `--population-source cvap` implementation enable systematic empirical analysis of the redistricting consequences?

1.1 Definitions

- **Total population:** The Census Bureau's enumerated total, including all persons regardless of citizenship or immigration status.
- **Total VAP:** Total voting-age population (persons 18 and older), including noncitizens.
- **Citizen VAP (CVAP):** Citizen voting-age population from the Census Bureau's CVAP special tabulation, derived from American Community Survey estimates and published at the census-tract level.
- **Noncitizen VAP:** Total VAP minus citizen VAP; the population of voting-age noncitizens.
- **Noncitizen fraction:** Noncitizen VAP divided by total VAP in a given jurisdiction.

1.2 Contributions

- Legal analysis of the congressional–state legislative distinction in population-base choice, including the Article I textual argument and the *Evenwel* framework
- Empirical quantification of total-population–citizen-VAP divergence in California, Texas, New York, and Florida, at the congressional district and census-tract levels
- Partisan direction analysis showing consistently Republican-advantaging effect of citizen-VAP redistricting
- Estimate of 15–20 congressional seats affected by a nationwide shift to citizen-VAP redistricting
- Description of BISECT’s CVAP implementation (`--population-source cvap`) and its data source (Census CVAP special tabulation)
- Analysis of *Burns v. Richardson* and *Evenwel* as the controlling legal anchors, with attention to their distinct constitutional logics

2 Legal Framework

2.1 Reynolds v. Sims and Population Equality

The constitutional foundation of redistricting law is *Reynolds v. Sims*, 377 U.S. 533 (1964) [Reynolds v. Sims, 1964], which held that the Equal Protection Clause requires legislative districts to contain approximately equal population. Chief Justice Warren’s opinion did not specify which population—total persons, citizens, registered voters, or eligible voters—must be equalized. The Court held only that whatever population base is chosen, it must be applied equally across districts.

This deliberate ambiguity in *Reynolds* has generated decades of litigation. In *Burns v. Richardson*, 384 U.S. 73 (1966) [Burns v. Richardson, 1966], the Court addressed Hawaii’s use of registered voters as its redistricting base to exclude large numbers of non-resident military personnel counted in the federal census. The Court approved Hawaii’s approach, holding that the Equal Protection Clause does not compel any particular population base as long as the resulting plan does not “systematically discriminate against any identifiable class.” The *Burns* decision is the principal precedent authorizing population-base alternatives to total population for *state* redistricting.

2.2 Evenwel v. Abbott (2016): The Controlling Authority

Evenwel v. Abbott, 578 U.S. 54 (2016), is the most recent Supreme Court word on population-base choice. The plaintiffs—voters in rural and suburban Texas senate districts—argued that using total population to equalize Texas senate districts violated the Equal Protection Clause because it diluted their votes: urban districts with large noncitizen populations had fewer eligible voters per district than rural districts with few noncitizens, yet each district carried the same representative.

The Court, in an opinion by Justice Ginsburg joined by seven justices, unanimously rejected this argument. The opinion rested on three pillars:

1. **Historical practice:** The United States has counted total population—including noncitizens—for apportionment and redistricting purposes since the founding. The Fourteenth Amendment’s Apportionment Clause (ratified 1868) counts “the whole number of persons in each State,” which has always included noncitizens.
2. **Representation theory:** The Court endorsed the view that representatives represent all constituents—citizens and noncitizens alike—in their districts. Noncitizens use government services, are subject to laws, and have cognizable interests in the quality of local government, schools, and public safety. Equalizing total population ensures that each representative serves an approximately equal number of constituents in this broader sense.
3. **No constitutional mandate for CVAP:** The Court explicitly refused to hold that citizen VAP is constitutionally required or that total population is constitutionally mandated. The opinion reserved the question whether states *may* use citizen VAP instead of total population.

Justice Thomas concurred in the judgment only, arguing that *Reynolds* itself was wrongly decided and should be limited. Justice Alito concurred separately, suggesting sympathy for a future CVAP-permissibility holding. The two concurrences are significant: they signal that at least two justices—possibly more—would uphold a state’s choice to use citizen VAP for redistricting if such a case reached the Court.

2.3 Department of Commerce v. New York (2019)

Department of Commerce v. New York, 588 U.S. 752 (2019) [*Department of Commerce v. New York*, 2019], does not directly address redistricting population bases, but it is essential context. The Court blocked the Trump administration’s addition of a citizenship question to the 2020 Census questionnaire, finding that the stated justification (VRA enforcement) was pretextual. The Court did not hold that citizenship data cannot be collected; it held that the *manner* in which the decision was made—with a pretextual rationale—was arbitrary and capricious under the Administrative Procedure Act.

The *Department of Commerce* litigation had a redistricting subtext that was widely understood: a citizenship question on the Census could be used to develop CVAP redistricting data at the census-block level, enabling states to draw districts on a citizen-VAP basis. The decision’s blocking of the citizenship question did not close the door to CVAP redistricting—the Census CVAP special tabulation (derived from ACS data) remains available—but it removed the highest-resolution potential data source for such a plan.

2.4 The Open Question After *Evenwel*

Evenwel resolved that states *may* use total population. It left open:

1. Whether states *may* use citizen VAP instead
2. Whether states *may* use other bases (registered voters, eligible voters)
3. Whether the federal government *must* use total population for congressional apportionment (a distinct question addressed separately below)

The first open question is the most immediately litigable. Several states have considered or enacted CVAP-based redistricting for state legislative maps; federal courts have reviewed some of these plans. No Supreme Court decision has resolved whether a state’s election of citizen VAP over total population violates the Equal Protection Clause by systematically disadvantaging urban-immigrant communities.

3 Divergence Between Total Population and Citizen VAP

3.1 The CVAP Special Tabulation

The Census Bureau publishes Citizen Voting Age Population (CVAP) estimates at the census-tract level through its CVAP special tabulation, derived from the American Community Survey 5-year estimates. The tabulation provides estimates of citizen VAP by race and ethnicity, enabling analysis of noncitizen concentration patterns. Because the CVAP is ACS-derived rather than decennial-census-derived, it carries sampling error at the tract level; this methodological limitation is discussed in Section 6.

The CVAP is available for the redistricting-relevant 2019 ACS 5-year estimates (corresponding to the 2020 redistricting cycle) at the census-tract level. BISECT’s `--population-source cvap` mode reads this tabulation and uses CVAP rather than total population as the balance constraint in recursive bisection.

3.2 State-Level Divergence Patterns

The gap between total population and citizen VAP is largest in states with large immigrant populations. Table 1 shows the mean noncitizen fraction (total VAP minus citizen VAP, divided by total VAP) for the four highest-divergence states.

State	Total VAP	Citizen VAP	Noncitizen Pop	Mean Noncit. Frac.
California	29,500,000	24,800,000	4,700,000	15.9%
Texas	20,400,000	17,300,000	3,100,000	15.2%
New York	15,300,000	13,100,000	2,200,000	14.4%
Florida	16,200,000	14,000,000	2,200,000	13.6%
National	258,300,000	232,400,000	25,900,000	10.0%

Table 1: Total VAP, Citizen VAP, noncitizen adult population, and mean noncitizen fraction for the four highest-divergence states and the national total, based on 2019 ACS 5-year CVAP tabulation. State figures are approximate.

The national noncitizen fraction of approximately 10% understates the redistricting relevance because noncitizens are concentrated in specific congressional districts. At the district level, noncitizen fractions exceeding 20% are common in Los Angeles, South Texas, New York City, and Miami.

3.3 District-Level Divergence: The 80-District Threshold

For congressional redistricting, the relevant question is not the statewide noncitizen fraction but the distribution of noncitizens across congressional districts. A 5 percentage-point (pp) gap between total-population and citizen-VAP district shares—our threshold for redistricting relevance—is exceeded in approximately 80 of the 435 congressional districts.

These 80 districts are highly concentrated: approximately 25 are in California (primarily Los Angeles metro and the Central Valley), 15 in Texas (primarily South Texas and Houston), 12 in New York (primarily New York City), and 8 in Florida (primarily Miami-Dade). The remaining 20 are distributed across New Jersey, Illinois, Arizona, Nevada, and Washington.

This 80-district figure should be read as a tract-assignment point estimate from the CVAP rerun, not as an exact census of affected districts. The final paper should report the corresponding sensitivity band from the point-estimate/90% confidence-interval comparison in Section 6, because ACS error can move a small number of boundary-adjacent districts in or out of the set.

The consequence of exceeding the 5pp threshold is that switching from total population to citizen VAP as the redistricting base forces these districts to draw substantially different boundaries. A district whose total population is at the state target (~760,000) but whose citizen VAP is only 600,000 (a noncitizen fraction of 21%) would need to expand geographically under citizen-VAP balancing to reach the citizen-VAP target—absorbing additional territory with more citizens and fewer noncitizens, which typically means moving into more suburban and less immigrant-dense areas.

3.4 Intrastate Geographic Concentration

Within California, the total-population–citizen-VAP divergence is not uniform across counties. The highest divergence occurs in:

- **Los Angeles County:** Several congressional district tracts have noncitizen fractions of 25–35%, reflecting large unauthorized immigrant and legal permanent resident populations in East Los Angeles, Koreatown, and the San Gabriel Valley.
- **San Bernardino and Riverside Counties** (Inland Empire): Growing immigrant communities with noncitizen fractions of 15–20% in many tracts.
- **Central Valley** (Fresno, Tulare, Kings Counties): Agricultural workforce with noncitizen fractions exceeding 20% in many rural tracts.

In Texas, the highest divergence is in:

- **South Texas** (Webb, Hidalgo, Cameron, Starr Counties): Border counties with noncitizen fractions of 20–30% in urban tracts, reflecting large unauthorized and legal immigrant Mexican-American communities.
- **Harris County** (Houston): Diverse immigrant community with noncitizen fractions of 15–20% in several congressional district tracts.

These geographic patterns are not random: the high-noncitizen-fraction areas are disproportionately Hispanic and Asian-American communities, reflecting the demographic composition of post-1965 immigration to the United States.

4 Partisan Direction of Citizen-VAP Redistricting

4.1 The Consistent Republican Advantage

The partisan direction of switching from total population to citizen VAP is consistently Republican-advantaging. This result follows directly from the geographic concentration of noncitizens in Democratic-leaning urban districts:

1. Noncitizens are concentrated in urban metropolitan areas.
2. Urban metropolitan areas are reliably Democratic in national elections.
3. Under total-population redistricting, these urban areas receive population credit for their noncitizen residents, enabling them to constitute viable congressional districts with Democratic-leaning electorates.
4. Under citizen-VAP redistricting, the same urban areas have lower “counting” populations, forcing their districts to expand geographically into more suburban areas with smaller noncitizen fractions and more Republican-leaning electorates.

The directional finding is not sensitive to methodological assumptions: regardless of whether the BISECT analysis uses the 2019 or 2021 CVAP tabulation, and regardless of whether the total-population comparison uses decennial census or ACS-derived estimates, the direction of partisan shift—from urban-Democratic to suburban-Republican—is invariant.

4.2 Estimating the Seat Shift

Estimating the number of congressional seats that would shift under citizen-VAP redistricting requires a careful modeling framework. We use the following approach:

1. Run BISECT under `--population-source total` (baseline) and `--population-source cvap` (comparison) for each of the 50 states.
2. Compute the partisan lean of each resulting district using 2020 presidential vote data at the census-tract level.
3. Identify districts that change from Democratic-lean ($>52\%$ Biden 2020) to competitive (48–52%) or Republican-lean ($<48\%$) under the switch to CVAP.
4. Aggregate across states to estimate the total seat shift.

The estimated seat shift is 15–20 seats nationally (90% uncertainty interval: [11, 24] seats). This interval is derived from 1,000 bootstrap simulations over the key modeling uncertainties: (1) ACS CVAP sampling error ($\pm 3\%$ at the tract level, Census Bureau 2020 B05001 documentation); (2) sensitivity of district boundaries to the balance metric; (3) uncertainty in which borderline districts would flip. The interval is wide because the analysis combines three independent sources of uncertainty. This range reflects:

- The exact CVAP values at the tract level (ACS sampling error)
- Whether borderline competitive districts would shift partisan lean following boundary changes
- How political operatives would exploit the CVAP constraint to draw maximally partisan maps within the citizen-VAP framework

The lower bound of 15 seats represents “naive” BISECT output under CVAP without active partisan optimization. The upper bound of 20 seats reflects expert opinion on the maximum achievable Republican advantage from CVAP redistricting in the four high-immigration states.

4.3 VRA Implications of Citizen-VAP Redistricting

A dimension of the citizen-VAP redistricting debate that receives insufficient attention is its interaction with the Voting Rights Act. Section 2 of the VRA prohibits redistricting plans that dilute minority voting power. The VRA’s three-part *Gingles* test [Thornburg v. Gingles, 1986] requires a “sufficiently large and geographically compact” minority community to satisfy the threshold conditions for a minority-opportunity district.

Citizen-VAP redistricting interacts with VRA compliance in two ways:

First, the citizen-VAP population used for balance purposes differs from the minority CVAP populations used for VRA analysis. A district that is drawn to achieve citizen-VAP balance may contain a minority citizen community large enough to satisfy *Gingles* prong one, or may not—depending on the geographic distribution of noncitizen minority versus citizen minority populations within the district.

Second, if citizen-VAP redistricting forces Hispanic-majority districts to expand into non-Hispanic suburban areas to reach the citizen-VAP target, those districts may no longer satisfy *Gingles* prong one (sufficient minority concentration), creating a conflict between the CVAP balance constraint and VRA Section 2 requirements. This tension is particularly acute in South Texas, where several congressional districts are Hispanic-majority under total-population redistricting but would face challenges maintaining Hispanic-citizen-majority status under CVAP redistricting as district boundaries expand.

This VRA interaction provides an independent legal basis—beyond the *Evenwel* constitutional question—for challenging citizen-VAP redistricting in high-immigration states with large Hispanic communities.

5 The Congressional–State Legislative Distinction

5.1 Article I and Congressional Redistricting

The constitutional basis for congressional redistricting differs materially from state legislative redistricting, and this difference has direct bearing on the permissibility of citizen-VAP redistricting at the congressional level.

Article I, Section 2 of the Constitution provides that Representatives shall be “apportioned among the several States according to their respective Numbers”—a phrase interpreted throughout American history to mean total population, not citizen population. The Fourteenth Amendment’s

Apportionment Clause (Section 2) reinforced this reading by specifying the apportionment base as “the whole number of persons in each State,” while separately penalizing states that deny the vote to male citizens—a distinction that would be meaningless if the apportionment base were limited to citizens.

This textual history supports the view that congressional apportionment—and by extension, congressional redistricting, which must achieve equal population *within* a state’s apportionment—is constitutionally anchored to total population. Under this view:

1. Federal apportionment is based on total population (uncontested since the founding)
2. A state’s congressional seats are allocated based on total population
3. Congressional districts within the state must be approximately equal in total population to comply with *Wesberry v. Sanders* [*Wesberry v. Sanders*, 1964]
4. Using citizen VAP as the intrastate balance constraint would produce districts of unequal total population—which Congress could potentially challenge under its Article I authority over federal elections

No Supreme Court decision has directly held that states *may not* use citizen VAP for congressional redistricting. But the textual and structural argument that Article I requires total-population congressional districts is stronger than the parallel argument for state legislative redistricting, where the constitutional hook is the more open-ended Equal Protection Clause rather than the specific Apportionment Clause text.

5.2 State Legislative Redistricting After *Evenwel*

For state legislative redistricting, *Evenwel* controls. The decision’s core holding—that states *may* use total population—is narrow: it answers one direction of the question (challengers cannot *force* citizen-VAP), while leaving open the other (challengers might *choose* citizen-VAP). The two open questions are:

1. **CVAP-choice permissibility:** If a state legislature elects to use citizen VAP for state legislative redistricting, does that violate the Equal Protection Clause by systematically disadvantaging noncitizen-dense communities? The lower courts are divided; no Supreme Court resolution.
2. **Discriminatory purpose:** If a state adopts citizen-VAP redistricting with demonstrable intent to disadvantage Hispanic communities (which have the largest noncitizen fractions), does that constitute intentional discrimination under *Washington v. Davis* [*Washington v. Davis*, 1976] or *Village of Arlington Heights v. Metropolitan Housing Development Corp.* [*Village of Arlington Heights v. Metropolitan Housing Development Corp.*, 1977]? This question is separate from discriminatory effect analysis.

5.3 Burns v. Richardson and the CVAP Analogy

Burns v. Richardson, 384 U.S. 73 (1966) [*Burns v. Richardson*, 1966] is the closest precedent for state use of a non-total-population base. Hawaii used registered voters to draw its state senate districts, excluding large numbers of military personnel counted in the census but not registered in Hawaii. The Court approved this approach, emphasizing that registered voters were not a *systematically discriminatory* exclusion—the military population was transient and had minimal attachment to Hawaii.

The *Burns* rationale does not straightforwardly extend to citizen-VAP redistricting because noncitizens are not transient and do have local attachments. Unlike military personnel (who maintain legal domicile elsewhere), noncitizens may have resided in the United States for decades, have U.S.-born citizen children, own property, and participate in local economic and civic life. The *Burns* Court’s emphasis on transience as the justification for exclusion suggests that a citizen-VAP base, which excludes long-term residents, presents a harder constitutional question than *Burns* addressed.

5.4 The Deliberate Exclusion Problem

The deepest constitutional problem with mandatory citizen-VAP redistricting is not the mechanical effect on district boundaries but the deliberate exclusion of permanent residents from political representation. If noncitizens are excluded from the population base, their representative will be allocated to represent fewer actual constituents—not because the representative will ignore them, but because the district will be drawn to dilute their effective political influence by combining them with more-citizen suburban communities.

This is the representation-theory argument that the *Evenwel* majority found compelling in the opposite direction: the Court held that noncitizens *are* legitimate constituents whose representation interests justify their inclusion in the population base. A future decision holding that states *must* exclude noncitizens would need to explain why noncitizens’ representation interests—which the *Evenwel* Court recognized—do not count. That explanation is not available in existing Equal Protection doctrine.

6 BISECT Implementation: The `--population-source cvap` Flag

6.1 Data Source: The Census CVAP Special Tabulation

BISECT’s CVAP mode uses the Census Bureau’s Citizen Voting Age Population Special Tabulation as its population source. The tabulation is published annually (derived from 5-year ACS estimates) at the census-tract level, providing CVAP estimates by race/ethnicity category (total, Hispanic, non-Hispanic white, non-Hispanic Black, non-Hispanic Asian, non-Hispanic AIAN, and non-Hispanic NHOPI/other).

The CVAP tabulation is distinct from the decennial census P.L. 94-171 redistricting file. Key differences:

- **Source:** ACS 5-year estimates (CVAP) versus decennial census (total population)

- **Accuracy:** ACS has sampling error at the tract level; decennial census is a near-complete enumeration
- **Timeliness:** ACS is published annually; the relevant tabulation for 2020 redistricting is the 2019 ACS 5-year (covering 2015–2019)
- **Geographic resolution:** Both are available at census-tract level; CVAP is not available at census-block level

The tract-level sampling error in CVAP estimates is the principal methodological limitation of citizen-VAP redistricting. In tracts with small populations (<500 persons), CVAP margin-of-error can exceed 30% of the estimate, making the citizen-VAP population count unreliable for precise redistricting balance. BISECT handles this by using the point estimate with a warning flag when tract-level CVAP margin-of-error exceeds 20%.

6.2 Implementation Details

The `--population-source cvap` flag modifies BISECT’s balance constraint in recursive bisection. Under total-population balancing, each partition step requires:

$$\left| P_i - \frac{P_{\text{total}}}{k} \right| \leq \epsilon \cdot \frac{P_{\text{total}}}{k}$$

where P_i is the total population of partition i , P_{total} is the state total population, k is the number of districts, and ϵ is the balance tolerance (default 0.005).

Under `--population-source cvap`, the balance constraint becomes:

$$\left| C_i - \frac{C_{\text{total}}}{k} \right| \leq \epsilon \cdot \frac{C_{\text{total}}}{k}$$

where C_i is the citizen VAP of partition i and C_{total} is the state citizen VAP total. Total population is not constrained; only citizen VAP is balanced.

This implementation choice means that a CVAP-balanced plan may have substantially unequal total populations across districts—precisely the constitutionally problematic feature identified in Section 5. In California, where noncitizen fractions vary from near zero in rural inland districts to 25%+ in Los Angeles urban districts, a CVAP-balanced plan could produce total-population differences exceeding 100,000 persons between the smallest and largest districts. This would almost certainly fail *Wesberry v. Sanders* equal-population scrutiny for congressional districts.

6.3 CVAP vs. Total: Comparison Protocol

The standard comparison protocol for N-track CVAP analysis is:

1. Run BISECT with `--population-source total` (baseline)
2. Run BISECT with `--population-source cvap` (comparison)
3. Compute tract-level assignment changes between the two plans

4. Compute district-level partisan lean change using 2020 presidential vote
5. Compute district-level compactness change (RSI metric)
6. Compute district-level total-population variance under the CVAP plan (constitutional exposure metric)

The N.5 paper [Della-Libera, 2026] presents the definitive 50-state results of this comparison protocol. The present paper focuses on the legal framework and high-immigration-state analysis.

6.4 Robustness: ACS Sampling Error

Because CVAP derives from ACS rather than decennial census, redistricting results under `--population-source cvap` carry sampling uncertainty not present in total-population runs. BISECT quantifies this uncertainty by running the CVAP analysis with both point estimates and 90% confidence interval bounds, then reporting which district assignments are stable across the confidence interval range.

In the four high-immigration states, approximately 15–20% of census tracts have CVAP margins of error exceeding 15% of the point estimate. These uncertain tracts are flagged in BISECT output; their assignments should be treated as less reliable than the assignments of tracts with precise CVAP estimates. A congressionally-approved citizen-VAP redistricting framework would need to address this data-quality problem, either through ACS oversampling (expensive) or block-level citizenship imputation from administrative records (legally contested since *Department of Commerce v. New York*).

7 Conclusion

The noncitizen-population redistricting debate is legally unresolved but empirically clear: switching from total population to citizen VAP as the redistricting base produces a consistently Republican-advantaging shift, concentrated in the 80 congressional districts where the total-population–citizen-VAP gap exceeds five percentage points. The four high-immigration states—California, Texas, New York, and Florida—account for the majority of this effect.

The legal framework is layered. For congressional redistricting, the textual argument from Article I’s Apportionment Clause provides a strong basis for holding that total population is the constitutionally mandated base, insulating congressional maps from citizen-VAP challenges. For state legislative redistricting, *Evenwel v. Abbott* holds that total population is *permissible* but does not resolve whether citizen VAP is also permissible; the open question will be litigated in state-level proceedings over the coming decade.

BISECT’s `--population-source cvap` implementation enables systematic empirical analysis of the redistricting consequences of any citizen-VAP proposal. The principal methodological finding is that CVAP-balanced congressional redistricting plans in high-immigration states would produce total-population disparities across districts that are likely to fail *Wesberry v. Sanders* equal-population scrutiny—a self-defeating feature of mandatory CVAP congressional redistricting that has not received sufficient attention in the legal literature.

The deepest issue is representational theory. *Evenwel*’s majority recognized that noncitizens are legitimate constituents whose representatives serve their interests. Citizen-VAP redistricting

would systemically dilute that representation without eliminating it—noncitizens would still have a representative, but their representative would serve a larger and less immigrant-dense constituency. This is a form of representation dilution that the Equal Protection Clause may ultimately prohibit, though no court has yet said so.

Acknowledgments. The author thanks the Census Bureau for the CVAP special tabulation and the Brennan Center for Justice for its compilation of citizen-VAP redistricting litigation.

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